

A Hessian–measure convexity index for nonsmooth functions

Full extension of the future problem in arXiv:1802.09630

Candidate solution packet — agent_lane_07

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Status. Candidate full solution, likely valid, pending expert review. The theorem extends the source’s Hessian/nuclear-norm index to nonsmooth functions with a finite matrix-valued distributional Hessian, including the stated min / max limits.

1 Source problem

Davydov, Moldavskaya, and Zitikis [1] define, for $h \in C^2(G)$, a global convexity index

$$\text{CONV}(h; G) = \frac{\int_G \text{tr}(H_h(x))_+ dx}{\int_G \|H_h(x)\|_* dx},$$

where H_h is the Hessian, A_+ is the spectral positive part of a symmetric matrix, and $\|\cdot\|_*$ is the nuclear norm. Their Theorem 3.1 identifies $\text{tr}(A_-)$ with the nuclear distance from A to the positive semidefinite cone. On page 15, Section 5, they ask for a modification that accommodates functions without second partial derivatives, specifically the limits

$$\min\{g(x), g(y)\}, \quad \max\{g(x), g(y)\}$$

of their power means as $\beta \rightarrow -\infty$ and $\beta \rightarrow +\infty$.

2 Idea of the proof

The ordinary Hessian is replaced by the distributional Hessian $M = D^2h$, a symmetric matrix-valued Radon measure. Write a polar density $M = A\rho$ against any positive control measure and take the spectral parts A_+ and A_- pointwise. This produces canonical positive-semidefinite matrix measures $M^+ = A_+\rho$ and $M^- = A_-\rho$. Integrating the source paper’s finite-dimensional matrix theorem proves that M^- is exactly the nuclear-total-variation distance from M to the cone of positive semidefinite matrix measures.

For a maximum or minimum of two smooth branches, the BV chain rule makes the new feature explicit. In addition to the ordinary Hessians on the two smooth regions, the switching hypersurface carries a rank-one measure

$$\pm |\nabla(u - v)| n \otimes n \mathcal{H}^{d-1}.$$

The sign is positive for max and negative for min. Thus the ridge has local index one for a maximum and zero for a minimum. Finally, the rank-one term is recovered as the concentration limit of the Hessians of the smooth power means, so the new index is the actual $\beta \rightarrow \pm\infty$ limit of the source’s indices.

5 Concluding notes

Theorem 3.1 is the backbone of this paper. It introduces an index for measuring lack of positive semidefiniteness in symmetric matrices. When specialized to the Hessian of sufficiently smooth functions, the theorem gives rise to an index suitable for measuring lack of convexity in functions, as discussed in Section 2. We have extensively illustrated these theoretical results in Section 4 with the help of a non-convex function, defined on a two-dimensional convex region, that arises naturally in applications.

Since our explorations of convexity, or lack of it, rely on the Hessian and its eigenvalues, the function under consideration must have second partial derivatives. Of course, this requirement may not always be fulfilled, like for example in the two limiting cases $\beta \downarrow -\infty$ and $\beta \uparrow \infty$ of the function h_β defined by equation (4.4), which give rise to non-differentiable functions $\min\{g(x), g(y)\}$ and $\max\{g(x), g(y)\}$, respectively. To accommodate such functions, the herein developed method requires a modification, which we posit as a future problem.

Figure 1: The future problem in the source paper, page 15, Section 5.

3 The measure-valued index

Let $G \subset \mathbb{R}^d$ be open. Denote by $\mathcal{M}(G; \text{Sym}_d)$ the finite symmetric matrix-valued Radon measures. Its nuclear total-variation norm is

$$\|M\|_{*,\mathcal{M}} := \sup \left\{ \int_G \Phi : dM : \Phi \in C_c(G; \text{Sym}_d), \sup_G \|\Phi\|_{\text{op}} \leq 1 \right\}.$$

If $M = A\rho$ for a positive control measure ρ , then $\|M\|_{*,\mathcal{M}} = \int_G \|A(x)\|_* d\rho(x)$. This is the $p = 1$ Hessian–Schatten variation used in the modern nonsmooth framework of [2]; see also the standard polar decomposition of vector measures in [3].

Let $\mathcal{P}(G)$ be the cone of positive-semidefinite matrix measures: a measure P belongs to $\mathcal{P}(G)$ when $z^T P z$ is a positive scalar measure for every $z \in \mathbb{R}^d$.

Theorem 3.1 (Measure-cone projection and nonsmooth convexity index). *Let $M \in \mathcal{M}(G; \text{Sym}_d)$. Choose a positive measure ρ with $M = A\rho$, and define*

$$M^+ = A_+\rho, \quad M^- = A_-\rho.$$

These measures do not depend on the choice of ρ , and

$$M = M^+ - M^-, \quad \|M\|_{*,\mathcal{M}} = \text{tr } M^+(G) + \text{tr } M^-(G). \quad (3.1)$$

Moreover,

$$\inf_{P \in \mathcal{P}(G)} \|M - P\|_{*,\mathcal{M}} = \|M^-\|_{*,\mathcal{M}} = \text{tr } M^-(G). \quad (3.2)$$

Consequently, if $h \in W_{\text{loc}}^{1,1}(G)$ and $D^2h \in \mathcal{M}(G; \text{Sym}_d)$, define

$$\text{LOC}_{\mathcal{M}}(h; G) = \text{tr}(D^2h)^-(G), \quad \text{CONV}_{\mathcal{M}}(h; G) = \frac{\text{tr}(D^2h)^+(G)}{\text{tr}(D^2h)^+(G) + \text{tr}(D^2h)^-(G)}. \quad (3.3)$$

When the denominator vanishes, h is affine on each connected component; we set $\text{CONV}_{\mathcal{M}} = 1$. Then:

1. $\text{CONV}_{\mathcal{M}}(h; G) \in [0, 1]$;
2. $\text{CONV}_{\mathcal{M}}(h; G) = 1$ if and only if h has a convex representative on convex G ;
3. $\text{CONV}_{\mathcal{M}}(h; G) = 0$ if and only if h has a concave representative and is not affine;
4. if $h \in W^{2,1}(G)$, formula (3.3) is exactly the source's global index, with $D^2h = H_h(x) dx$.

Proof. Changing the control measure multiplies the density by a nonnegative scalar, and spectral positive and negative parts are positively homogeneous. Hence $M^\pm$ are intrinsic. The identities in (3.1) follow pointwise from $A = A_+ - A_-$ and $\|A\|_* = \text{tr } A_+ + \text{tr } A_-$.

Fix $P \in \mathcal{P}(G)$, and dominate both M and P by $\sigma = |M|_* + \text{tr } P$. Write $M = B\sigma$ and $P = C\sigma$. Then $C(x) \succeq 0$ for σ -almost every x . The source paper's finite-dimensional theorem, applied pointwise, yields

$$\|B(x) - C(x)\|_* \geq \text{tr } B_-(x).$$

After integration,

$$\|M - P\|_{*,\mathcal{M}} \geq \int_G \text{tr } B_- d\sigma = \text{tr } M^-(G).$$

Equality is attained by $P = M^+$, because $M - M^+ = -M^-$. This proves (3.2) and (3.3).

For the calibration statement, the classical distributional characterization says that a locally integrable function on a convex open set is convex exactly when its distributional Hessian is positive semidefinite. One quick proof is to mollify: a positive-semidefinite distributional Hessian gives convex smooth mollifications on interior subdomains, and their local L^1 limit has a convex representative. The converse follows by restricting a convex function to lines and taking nonnegative second directional derivatives. Thus $(D^2h)^- = 0$ is equivalent to convexity. Applying the same argument to $-h$ proves the concave statement. The $W^{2,1}$ specialization is immediate from the absolutely continuous density $H_h(x)$. $\square$

The pointwise index also has a precise measure analogue. With

$$\kappa_h = \text{tr}(D^2h)^+ + \text{tr}(D^2h)^-,$$

set

$$\text{CONV}_{\mathcal{M}}(h, x) = \frac{d \text{tr}(D^2h)^+}{d\kappa_h}(x) \quad \text{for } \kappa_h\text{-a.e. } x. \quad (3.4)$$

At Lebesgue points of a classical Hessian this is the source's eigenvalue ratio. Formula (3.4) remains meaningful on singular ridges, where Lebesgue measure sees nothing.

4 Exact formulas for maxima and minima

Theorem 4.1 (Switching-interface formula). *Let $u, v \in C^2$ on a neighborhood of $\overline{G}$, put $\phi = u - v$, and suppose 0 is a regular value of ϕ . Let*

$$\Sigma = G \cap \{u = v\}, \quad n = \frac{\nabla \phi}{|\nabla \phi|}.$$

Then, as symmetric matrix-valued measures,

$$D^2 \max\{u, v\} = \mathbf{1}_{\{u > v\}} D^2 u \, dx + \mathbf{1}_{\{v > u\}} D^2 v \, dx + |\nabla \phi| \, n \otimes n \, \mathcal{H}^{d-1} \upharpoonright \Sigma, \quad (4.1)$$

$$D^2 \min\{u, v\} = \mathbf{1}_{\{u < v\}} D^2 u \, dx + \mathbf{1}_{\{v < u\}} D^2 v \, dx - |\nabla \phi| \, n \otimes n \, \mathcal{H}^{d-1} \upharpoonright \Sigma. \quad (4.2)$$

The singular term has nuclear mass

$$J_\Sigma = \int_\Sigma |\nabla(u - v)| \, d\mathcal{H}^{d-1}. \quad (4.3)$$

It contributes J_Σ entirely to the positive curvature mass of the maximum and entirely to the negative curvature mass of the minimum. Consequently (3.3) gives explicit indices by adding (4.3) to the appropriate bulk spectral mass.

Proof. Since $\max\{u, v\} = v + \phi_+$, the BV chain rule and the Gauss–Green formula give

$$\nabla \max\{u, v\} = \nabla v + \mathbf{1}_{\{\phi > 0\}} \nabla \phi$$

and

$$D\mathbf{1}_{\{\phi > 0\}} = \frac{\nabla \phi}{|\nabla \phi|} \, \mathcal{H}^{d-1} \upharpoonright \Sigma.$$

Differentiating once more produces (4.1), because $\nabla \phi \otimes \nabla \phi / |\nabla \phi| = |\nabla \phi| n \otimes n$. The minimum identity follows either in the same way or from $\min\{u, v\} = u + v - \max\{u, v\}$. The rank-one matrix $n \otimes n$ is positive semidefinite and has nuclear norm one, proving (4.3) and the sign assertions. $\square$

The regular-value assumption is a transparent generic hypothesis. The same formula holds whenever the switching set is countably rectifiable and its critical part has $\mathcal{H}^{d-1}$ -measure zero, which covers isolated critical crossings by approximation.

Corollary 4.2 (Closed one-dimensional branch formulas). *Let $I = (a, b)$, let $g \in C^2(I)$ be strictly monotone with $g' \neq 0$, and set $Q = I^2$. Define*

$$J = 2 \int_a^b |g'(t)| \, dt,$$

and

$$W_{\max}(t) = \begin{cases} t - a, & g \text{ increasing,} \\ b - t, & g \text{ decreasing,} \end{cases} \quad W_{\min}(t) = b - a - W_{\max}(t).$$

For $h_{\max}(x, y) = \max\{g(x), g(y)\}$, its positive and negative masses are

$$P_{\max} = 2 \int_a^b W_{\max}(t) (g''(t))_+ \, dt + J,$$

$$N_{\max} = 2 \int_a^b W_{\max}(t) (g''(t))_- \, dt.$$

For $h_{\min}(x, y) = \min\{g(x), g(y)\}$, they are

$$P_{\min} = 2 \int_a^b W_{\min}(t) (g''(t))_+ \, dt,$$

$$N_{\min} = 2 \int_a^b W_{\min}(t) (g''(t))_- \, dt + J.$$

In both cases $\text{CONV}_{\mathcal{M}} = P/(P + N)$.

Proof. Strict monotonicity makes Σ the diagonal. Along it, $|\nabla(g(x) - g(y))| = \sqrt{2}|g'(t)|$ and $d\mathcal{H}^1 = \sqrt{2} \, dt$, yielding J . Integrating the selected diagonal Hessian $\text{diag}(g''(x), 0)$ or $\text{diag}(0, g''(y))$ over the two triangles gives the weights shown above. $\square$

5 Recovery as the smooth power-mean limit

Theorem 5.1 (No large- $|\beta|$ surrogate is needed). *Assume the hypotheses of Theorem 4.1, with $u, v \geq m > 0$, and assume $\mathcal{H}^{d-1}(\Sigma \cap \partial G) = 0$. For $\beta \neq 0$, put*

$$h_\beta = \left(\frac{u^\beta + v^\beta}{2} \right)^{1/\beta}.$$

Then on G , in the weak-star sense of matrix measures and strictly in nuclear total variation,

$$\begin{aligned} D^2 h_\beta dx &\longrightarrow D^2 \max\{u, v\} & (\beta \rightarrow +\infty), \\ D^2 h_\beta dx &\longrightarrow D^2 \min\{u, v\} & (\beta \rightarrow -\infty). \end{aligned}$$

The positive and negative spectral masses converge separately. Hence, whenever the limiting denominator is nonzero,

$$\text{CONV}(h_\beta; G) \longrightarrow \begin{cases} \text{CONV}_{\mathcal{M}}(\max\{u, v\}; G), & \beta \rightarrow +\infty, \\ \text{CONV}_{\mathcal{M}}(\min\{u, v\}; G), & \beta \rightarrow -\infty. \end{cases} \quad (5.1)$$

Proof. Write $a = \log u$, $b = \log v$, and

$$p_\beta = \frac{e^{\beta a}}{e^{\beta a} + e^{\beta b}}, \quad m_\beta = p_\beta \nabla a + (1 - p_\beta) \nabla b.$$

A direct differentiation gives

$$\begin{aligned} D^2 h_\beta &= h_\beta [m_\beta \otimes m_\beta + p_\beta D^2 a + (1 - p_\beta) D^2 b \\ &\quad + \beta p_\beta (1 - p_\beta) (\nabla a - \nabla b) \otimes (\nabla a - \nabla b)]. \end{aligned} \quad (5.2)$$

Away from Σ , the first line converges to the Hessian of the selected branch. In the second line, $\beta p_\beta (1 - p_\beta)$ is the derivative of the logistic function in the normal variable $a - b$, with integral one. The coarea formula therefore gives, for $\beta \rightarrow +\infty$,

$$\beta h_\beta p_\beta (1 - p_\beta) (\nabla a - \nabla b)^{\otimes 2} dx \xrightarrow{*} |\nabla(u - v)| n \otimes n \mathcal{H}^{d-1} \upharpoonright \Sigma.$$

Indeed, on Σ , where $u = v = s$, one has $s|\nabla(a - b)| = |\nabla(u - v)|$. For $\beta \rightarrow -\infty$ the same logistic layer has negative sign and selects the minimum.

To obtain strict nuclear convergence, split G into a thin tubular neighborhood of Σ and its complement. On the complement, dominated convergence applies to (5.2). In the tube, the first line is uniformly bounded, while the rank-one last line has the indicated sign and its nuclear mass converges by coarea to (4.3). The integral of the bounded term over the tube vanishes with its width. This proves convergence of total nuclear mass. For symmetric matrices,

$$\text{tr } A_+ = \frac{1}{2}(\|A\|_* + \text{tr } A), \quad \text{tr } A_- = \frac{1}{2}(\|A\|_* - \text{tr } A),$$

so nuclear convergence together with weak convergence of the trace measures gives separate convergence of the positive and negative masses. Formula (5.1) follows. $\square$

6 The source paper’s concrete function

For

$$g(t) = \frac{1}{2}(1-t)^2 + 0.01t^{1/4},$$

one has

$$g'(t) = t - 1 + 0.0025t^{-3/4}, \quad g''(t) = 1 - 0.001875t^{-7/4}.$$

On $I = [0.1, 0.9]$, g is strictly decreasing and strictly convex. Corollary 4.2 gives

$$J = 0.7917667515752844, \quad qquad P_{\min} = 2 \int_{0.1}^{0.9} (t - 0.1)g''(t) dt = 0.6360956571149691,$$

and therefore

$$\text{CONV}_{\mathcal{M}}(\min\{g(x), g(y)\}; I^2) = 0.4454880619053803.$$

The maximum is convex on this square, so its index is exactly one. A midpoint quadrature of the classical Hessian on an 800×800 grid gives 0.4462763224 at $\beta = -100$ and 1.0000000000 at $\beta = 100$, consistent with Theorem 5.1. The reproducible checker is `code/verify_formulas.py`.

The source’s g is not globally monotone: it has two critical points, and g'' is singular at the endpoint zero. The general switching formula (4.1)–(4.3) handles every regular diagonal or off-diagonal equality branch. On the full open unit square the Hessian measure is locally finite; global finite indices should be taken on compact interior subdomains (as the source does for its expanding squares) or defined by exhaustion when the limiting ratio exists.

7 Verification, novelty, and review focus

Checks. The script tests the finite-dimensional distance inequality against 2,000 random symmetric/PSD matrix pairs, verifies equality at the spectral positive part, evaluates the closed interface formula, and compares it with smooth power means. These computations are sanity checks, not ingredients of the proof.

Bounded novelty search. On 11 August 2026 we searched the exact source title and future-problem terms, as well as “distributional Hessian matrix-valued measure distance positive semidefinite cone nuclear norm”, “Hessian-Schatten total variation nonsmooth convexity”, and close variants. We found the later Hessian–Schatten framework of Aziznejad–Campos–Unser [2] and related 2023–2024 work, but no paper stating the signed PSD-measure projection identity (3.2), the resulting convexity index, the min / max ridge formulas, or the power-mean index limit (5.1). Novelty is therefore plausible, not certified. The framework itself is known; the signed convexity construction and its application to the source problem are the candidate contribution.

Human-review recommendation. Promote for expert review as a candidate full solution. The most important checks are: (i) the intrinsic spectral decomposition of a matrix measure; (ii) strict nuclear convergence in Theorem 5.1, especially if the regular-value hypothesis is weakened; and (iii) treatment by exhaustion near the source function’s endpoint singularity. None of these affects the exact interior formulas under the stated hypotheses.

References

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- [2] S. Aziznejad, J. Campos, and M. Unser, *Measuring complexity of learning schemes using Hessian–Schatten total variation*, arXiv:2112.06209; SIAM J. Math. Data Sci. 5 (2023), 422–445.
- [3] L. Ambrosio, N. Fusco, and D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems*, Oxford Mathematical Monographs, 2000.